

Exercise 1

Consider the ARMA process:

$$Y_t = 0.5Y_{t-1} + \varepsilon_t - 0.7\varepsilon_{t-1} + 0.1\varepsilon_{t-2},$$

where ε_t is white noise. Determine if Y_t is a regular ARMA process. Convert it to a regular one if it isn't.

Correction: The process is causal if the magnitudes of all roots of the AR polynomial are strictly larger than 1. Since the root of $\phi(B) = 1 - .5B$ is 2, we deduce that the ARMA process is causal. The process is invertible if the magnitudes of all roots of the MA polynomial are strictly larger than 1. Since the roots of $\theta(B) = 1 - .7B + .1B^2 = (1 - .5B)(1 - .2B)$ are 2 and 5 respectively, we deduce that the process is invertible. The process is regular if in addition to being causal and invertible, the AR and MA polynomials have no common roots. Thus, because 2 is a common root to both $\phi(B)$ and $\theta(B)$, we deduce that the ARMA process is not regular. To convert the process to a regular one, we cancel the common factor in

$$\phi(B)Y_t = \theta(B)\varepsilon_t.$$

This leads to

$$Y_t = (1 - 0.2B)\varepsilon_t = \varepsilon_t - 0.2\varepsilon_{t-1},$$

which is an invertible MA(1) process. □

Exercise 2

Consider the process

$$Y_t = 1.3Y_{t-1} - 0.3Y_{t-2} + \varepsilon_t - \theta\varepsilon_{t-1},$$

where ε_t is white noise.

1. Assuming $\theta = 0$, identify the model using the ARIMA(p, d, q) notation and determine whether Y_t is stationary. If it is not, find a transformation that makes it stationary.
2. Assuming $|\theta| < 1$, with $\theta \neq 0$, determine if the process is invertible. If it is, determine the first three coefficients π_0, π_1, π_2 in the representation $\varepsilon_t = \sum_{i=0}^{\infty} \pi_i Y_{t-i}$.
3. Assuming $\theta = 0$, give a recursive formula for the Box-Jenkins h -step-ahead forecast Y_{n+h}^n in terms of Y_{n+h-1}^n and Y_{n+h-2}^n , for $h > 2$. Show all steps and justify your reasoning.

Correction:

1. The process has a unit root which can be seen either from the characteristic equation of the AR polynomial or otherwise. Thus, it is not stationary. The transformation $\nabla Y_t = 0.3\nabla Y_{t-1} + \varepsilon_t$ leads to a causal AR(1) process, which is stationary. Thus, the process can be identified as an ARIMA(1, 1, 0)

2. Since $|\theta| < 1$ we have that the root of $\theta(B) = 1 - \theta B$ is $|1/\theta| > 1$. Hence, the process is invertible. Inverting the process gives

$$\begin{aligned}
\varepsilon_t &= \sum_{i=0}^{\infty} \theta^i (1 - 0.3B) \nabla Y_{t-i} = \sum_{i=0}^{\infty} \theta^i (1 - 0.3B)(1 - B)Y_{t-i} \\
&= \sum_{i=0}^{\infty} \theta^i (1 - 1.3B + 0.3B^2)Y_{t-i} \\
&= \sum_{i=0}^{\infty} \theta^i [Y_{t-i} - 1.3Y_{t-i-1} + 0.3Y_{t-i-2}] \\
&= Y_t - 1.3Y_{t-1} + 0.3Y_{t-2} + \theta[Y_{t-1} - 1.3Y_{t-2} + 0.3Y_{t-3}] + \theta^2[Y_{t-2} - 1.3Y_{t-3} + 0.3Y_{t-4}] \\
&\quad + \dots
\end{aligned}$$

Reorganizing gives $\pi_0 = 1, \pi_1 = \theta - 1.3$ and $\pi_2 = 0.3 - 1.3\theta + \theta^2$.

3. Starting from $h = 1$, we have

$$Y_{n+1} = 1.3Y_n - 0.3Y_{n-1} + \varepsilon_{n+1}.$$

Setting ε_{n+1} to zero gives $Y_{n+1}^n = 1.3Y_n - 0.3Y_{n-1}$. Likewise, for $h = 2$ we have

$$\begin{aligned}
Y_{n+2} &= 1.3Y_{n+1} - 0.3Y_n + \varepsilon_{n+2} \\
&= 1.3(1.3Y_n - 0.3Y_{n-1}) - 0.3Y_n + \varepsilon_{n+1} + \varepsilon_{n+2}
\end{aligned}$$

Setting ε_{n+1} and ε_{n+2} to zero gives $Y_{n+2}^n = 1.3Y_{n+1}^n - 0.3Y_n$. Iterating the argument shows that for $h > 2$,

$$Y_{n+h}^n = 1.3Y_{n+h-1}^n - 0.3Y_{n+h-2}^n.$$

□

Exercise 3

Consider the EGARCH(1,1) model $Z_t = \sigma_t \varepsilon_t$ where

$$\log \sigma_t^2 = c + \alpha |Z_{t-1}| + \gamma Z_{t-1} + \beta \log \sigma_{t-1}^2,$$

and $\varepsilon_t \sim (0, 1)$.

1. Explain how the term γZ_{t-1} captures the leverage effect, where negative shocks have a larger impact on volatility than positive shocks of the same magnitude.
2. Derive a formula for forecasting the next day-ahead conditional variance σ_{n+1}^2 using only $Z_n, Z_{n-1}, \dots$, and the parameters c, α, β and γ , assuming $|\beta| < 1$. Explain all steps involved in calculating this forecast.

Correction:

1. The term $\alpha|Z_{t-1}|$ captures the magnitude effect of past shocks, representing the symmetric response of volatility to the absolute value of the shock. Regardless of whether the shock is positive or negative, its absolute magnitude contributes to volatility.

The leverage effect is a well-documented phenomenon in financial markets, where negative shocks (e.g., bad news) tend to increase volatility more than positive shocks (e.g., good news) of equal magnitude. The term γZ_{t-1} is specifically designed to capture this effect, representing the asymmetric response of volatility to the sign of the shock. That is, if γ is negative, negative shocks will have a larger impact on volatility than positive shocks of the same magnitude and vice versa.

2. Using the backshift B , we rewrite the logarithm of the conditional variance as

$$\log(\sigma_t^2) = c + \alpha Z_{t-1} + \gamma|Z_{t-1}| + \beta B \log(\sigma_t^2)$$

Thus

$$\begin{aligned} \log(\sigma_t^2) &= \sum_{k=0}^{\infty} \beta^k (c + \alpha Z_{t-1-k} + \gamma|Z_{t-1-k}|) \\ \sigma_{n+1}^2 &= \exp \left\{ \sum_{k=0}^{\infty} \beta^k (c + \alpha Z_{n-k} + \gamma|Z_{n-k}|) \right\} \end{aligned}$$

□

Exercise 4

Let us consider the equation, with Z_t a white noise:

$$X_t = -\frac{1}{6}(X_{t-1} - 4X_{t-2} + X_{t-3}) + Z_t, \forall t \in \mathbb{Z}.$$

We want to show that there is no stationary solution to this equation. To that purpose, we proceed by *reductio ad absurdum*. We assume there exist a stationary solution X . Let ν_X be its spectral density.

1. Let $\Phi(z) = 1 + \frac{1}{6}(z - 4z^2 + z^3)$, show that for any continuous 2π periodical function $f : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$\int_0^{2\pi} f(u) |\Phi(e^{iu})|^2 \nu_X(du) = \int_0^{2\pi} f(u) \frac{1}{2\pi} du.$$

2. (harder question) For $\delta > 1$, let $f_\delta(u) = 1/(\delta + \cos(u))$. Show that the function $u \mapsto f_\delta(u) |\Phi(e^{iu})|^2$ is bounded independently of δ on $\mathbb{R}$ and find a contradiction.

Correction:

1. Let $\beta \in \ell^1(\mathbb{Z})$ be defined by $\beta_0 = 1$, $\beta_1 = -1/6$, $\beta_2 = 2/3$, $\beta_3 = -1/6$ and $\beta_k = 0$ otherwise. Then, $\Phi(z) = \sum_{k \in \mathbb{Z}} \beta_k z^k$ for all $z \in \mathbb{C}$, and thus $F_\beta X = Z$. By one of the note's theorem, we have that the spectral densities of X and Z are linked by:

$$|\Phi(e^{iu})|^2 \nu_X(du) = \nu_Z(du).$$

Thus, for any continuous, 2π periodical function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$\int_0^{2\pi} f(u) |\Phi(e^{iu})|^2 \nu_X(du) = \int_0^{2\pi} f(u) \nu_Z(du).$$

But, one of the other course's theorem states that as Z is a white noise $(0, 1)$, it has a constant spectral density (wrt Lebesgue measure) $\mu_Z(du) = 1/(2\pi)du$. Thys, for any any continuous, 2π periodical function $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$\int_0^{2\pi} f(u) |\Phi(e^{iu})|^2 \nu_X(du) = \int_0^{2\pi} f(u) 1/(2\pi)(du).$$

2. Note that $\Phi(z) = (1 + z)(1 - \frac{5}{6}z + \frac{1}{6}z^2)$, so that,

$$f_\delta(u) |\Phi(e^{iu})|^2 = f_\delta(u) |1 + e^{iu}|^2 \left| 1 - \frac{5}{6}e^{iu} + \frac{1}{6}e^{2iu} \right|^2 = \frac{(1 + \cos(u))^2 + \sin^2(u)}{\delta + \cos(u)} g(u),$$

where $g(u) = \left| 1 - \frac{5}{6}e^{iu} + \frac{1}{6}e^{2iu} \right|^2$ is a 2π periodical continuous function (thus bounded independently of δ). We have:

$$0 \leq \frac{(1 + \cos(u))^2 + \sin^2(u)}{\delta + \cos(u)} \leq 1 + \cos(u) + (1 - \cos(u)) \frac{1 + \cos(u)}{\delta + \cos(u)} \leq 2.$$

Then, as $\delta + \cos(u) \geq 1 + \cos(u) \geq 0$, we deduce that $u \mapsto f_\delta(u) |\Phi(e^{iu})|^2$ is bounded on $[-1, 1]$, uniformly on $\delta > 1$. Thus,

$$\int_0^{2\pi} \frac{1}{2\pi(\delta + \cos(u))} du = \int_0^{2\pi} f_\delta(u) |\Phi(e^{iu})|^2 \mu_X(du)$$

is bounded independently of $\delta > 1$, which is impossible as by monotonous convergence and when $\delta \rightarrow 1^+$, this would imply that

$$\int_0^{2\pi} \frac{1}{2\pi(1 + \cos(u))} du$$

is a convergent integral.

□